

HARSH KUMAR JHA

B.TECH (2028) IN ARTIFICIAL INTELLIGENCE

+91 8368805074

harshjha24110130@gmail.com

LinkedIn

Codeforces

GitHub

EDUCATION

Indian Institute of Technology Gandhinagar

B.Tech. in Artificial Intelligence

CPI: 8.51

2024–Present

New Green Field School

Class XII (Physics, Chemistry, Maths)

Score: 87.4%

2021–2023

Cambridge International School

Class X

Score: 93.5%

2021

SCHOLASTIC ACHIEVEMENTS & AWARDS

- **Competitive Programming:** Expert on Codeforces (1609 rating) and 3★ on CodeChef (1603 rating) : strong algorithmic problem-solving.
- **Selected for Amazon ML Summer School 2026:** Selected among the top ~2% (3,000 from 1.5 lakh+ applicants) for an advanced machine learning program conducted by Amazon data scientists.
- **Selected as a BIIC (Barclays) Mentee** for the June–July 2026 mentorship program through a competitive selection process.
- **Department change** from Civil Engineering to Artificial Intelligence at IIT Gandhinagar : awarded on merit and performance.
- **Department Rank 1** in First Semester at IIT Gandhinagar.

RESEARCH EXPERIENCE

Reimagining Point Cloud Denoising in Latent Space

SUPERVISOR: PROF. SHANMUGANATHAN RAMAN | [GitHub](#)

[Aug '25 – Dec '25]

- Built an **end-to-end latent-space diffusion framework** for 3D point cloud denoising using geometry-aware autoencoders to preserve global shape structure.
- Designed hierarchical **transformer** encoders with **self-attention** and timestep conditioning for multi-scale diffusion, outperforming raw coordinate-space diffusion.
- Improved **Chamfer Distance** by 2× over diffusion baselines (1.23 vs 2.28 at 1% noise, 10k points), validated via ablations.

SELECTED PROJECTS

Semantic Search Engine: Dense Retrieval & Hybrid Ranking

SUPERVISOR: PROF. ANIRBAN DASGUPTA | [Project Link](#)

[Jan '26 – Apr '26]

- Benchmarked 7 dense & sparse **retrieval** strategies (BM25, Word2Vec, BERT bi-encoder, cross-encoder) on SciFact (BEIR); hybrid BM25+BERT fusion beat cross-encoder reranking on precision (0.0923 vs 0.0917) at 8× **lower latency** (31ms vs 240ms).
- Built a custom **HNSW** nearest-neighbour index from scratch (C++, pybind11) for cosine-similarity search, reaching 98.2% top-10 overlap with FAISS exhaustive search.

Beyond GANSpace: Interpretable Control in GAN Latent Spaces

StyleGAN2, GAN, PCA, CLIP, PyTorch

[Jan '26 – Apr '26]

- Reproduced **GANSpace** on StyleGAN2-FFHQ, using **PCA** on W -space latents to extract interpretable edit directions, ranked via **CLIP** contrastive scoring.
- Found PCA directions semantically entangled (e.g., “beard” axis behaved as gender axis), traced to attribute correlations in training data.

TECHNICAL SKILLS

- **Languages & Fundamentals:** C++, Python, Data Structures & Algorithms, Object-Oriented Programming
- **ML / AI:** Machine Learning, Deep Learning, Diffusion Models, Transformers, GANs
- **Libraries:** PyTorch, NumPy, Pandas, Matplotlib, Optuna